Education search and background extraction

Literature search and structured evidence extraction on **AI for Social Good (AI4SG) tools in the education sector** in **Bangladesh, India, Indonesia and Thailand**.

1. Objective

Identify relevant empirical studies, evaluations, conference papers, working papers, government reports, and credible institutional reports on AI-enabled agricultural tools **that have been actually tested, deployed, implemented, or used in the education sector** in Bangladesh, India, Indonesia, or Thailand.

Geographic scope: a source is eligible if the AI tool was tested or used in at least one of the four countries. Multi-country studies are eligible only if results, sample sizes, or implementation details for at least one of the four countries are reported separately; extract only the country-specific information.

The purpose is to extract factual information that can later be used for analysis of:

- intervention
- adoption
- effectiveness
- implementation
- deployment
- scaling
- cost
- financing

Focus on collecting accurate and comparable information across studies.

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2. Definition of AI4SG in education

For this review, define **AI4SG in education** as:

> The use of AI-enabled systems to address socially important educational problems by improving learning, access, equity, inclusion, teaching capacity, or the efficiency and quality of education delivery, especially where students, teachers, schools, or education systems face resource, capacity, or access constraints.

Do not require the paper itself to use the term "AI for Social Good."

Include an application when:

1. AI is a substantive component of the tool or service;
2. the application addresses an educational problem; and
3. the intended outcome concerns learning, access, equity, inclusion, teaching capacity, affordability, or education-system performance.

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3. AI applications to include

Include, where relevant:

- intelligent tutoring systems
- personalized adaptive learning
- adaptive learning platforms
- generative-AI or LLM tutors
- educational chatbots
- teacher copilots
- AI-assisted teacher support
- automated feedback
- automated assessment
- AI-supported formative assessment
- dropout or learning-risk prediction
- early-warning systems
- AI for accessibility or special education
- AI translation or language-learning systems
- AI-supported education administration
- AI-supported public-service delivery in education

Older systems may be included if adaptive algorithms, learner modelling, prediction, NLP, automated personalization, or automated feedback are substantive components.

Exclude ordinary digital education tools where AI is not a substantive component.

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4. Type of evidence to prioritize

Prioritize sources that contain information on **real interventions or real deployments**, including:

- randomized trials
- field experiments

- quasi-experimental evaluations
- implementation studies
- deployment evaluations
- scaling studies
- observational studies using actual usage data
- government evaluations
- international-organization reports
- credible program evaluations
- implementation reports
- cost or financing reports

Do not prioritize:

- opinion pieces
- purely conceptual papers
- general discussions of AI in education
- attitude-only studies
- intention-to-use surveys without actual usage data
- purely technical machine-learning papers without real educational deployment
- papers where AI is used only for generic convenience

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5. Search approach

Search broadly using combinations of terms such as:

AI terms

"artificial intelligence"
"generative AI"
"large language model"
"AI tutor"
"intelligent tutoring system"
"adaptive learning"
"personalized learning"
"personalized adaptive learning"
"educational chatbot"
"teacher copilot"
"automated feedback"
"automated assessment"
"early warning system"
"learning analytics"
"machine learning"

Education terms

education
school
student
teacher
classroom
learning
primary school
secondary school
higher education
public school

Evidence terms

intervention
trial
experiment
evaluation
implementation
adoption
usage
deployment
scale
scaling
cost
cost-effectiveness
financing
procurement
government adoption

For highly relevant programs, also search the **program or tool name directly** to locate additional studies or implementation reports.

Where possible, connect multiple sources referring to the same AI tool or program.

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6. Unit of Analysis and Cross-Source Linking

Use the AI intervention or deployment as the main unit of analysis, rather than the individual paper or source.

A single AI tool may be used in multiple interventions or deployments, while multiple sources may describe the same intervention or deployment. Assign unique IDs to identify these entities and link related evidence across sources.

Assign three types of IDs:

- Tool ID: Identifies a distinct AI tool or system. Retain the same Tool ID when the tool is used across different interventions or deployments.
- Intervention ID: Identifies a distinct intervention or real-world deployment of an AI tool. Assign a new ID when the intervention design, intended application, or deployment setting differs materially.
- Source ID: Identifies each distinct paper, report, government document, webpage, or other source.

Cross-source identification and linking:

- Before assigning a new ID, check whether the AI tool or intervention has already been identified.
- Use tool/programme names, providers, implementing organizations, locations, implementation periods, target populations, and intervention designs to identify related records.
- When multiple sources describe the same intervention or deployment, retain the same Intervention ID and link all relevant Source IDs.
- When the same AI tool is used in different interventions or deployments, assign separate Intervention IDs while retaining the common Tool ID.
- Combine complementary information across related sources, including effectiveness, adoption, costs, implementation, financing, and scaling, while preserving the original source of each finding.
- Do not double-count the same intervention or outcome simply because it appears in multiple sources. Check whether different publications report the same underlying evaluation, sample, or results.
- If it is unclear whether two sources describe the same intervention, flag the relationship for verification rather than automatically merging the records.

Recording rule: Maintain one main extraction record per distinct intervention/deployment, linked to its Tool ID, Intervention ID, and all relevant Source IDs.

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7. Information to extract

For every relevant intervention, extract only information that is directly useful for later analysis.

A. Source information

- Full citation
- Authors
- Year
- Paper/report title
- Journal or institution
- DOI
- Official URL
- Publication type

B. Context

- Country
- Region or city, if reported
- Country income level (World Bank classification for the study year, if the study year is not specified, use classification for the current year, and with a note)
- Urban / rural / mixed
- Public / private education setting
- Education level
- Subject area
- Target population
- Age or grade
- Number of students
- Number of teachers
- Number of schools
- Other relevant sample characteristics

C. Educational problem

Record the main problem the AI tool is intended to address.

Examples:

- low learning outcomes
- students below grade level
- teacher shortage
- limited teacher expertise
- high student-teacher ratio
- lack of individualized feedback
- dropout
- low attendance
- limited educational access
- language barriers
- disability-related barriers

- weak assessment capacity
- administrative inefficiency

Use the authors' description where possible.

D. AI application

Extract:

- AI tool / program name
- Provider / developer
- AI application category
- Technology type
- Main function of the AI
- Primary user:
    - student
    - teacher
    - tutor
    - administrator
    - policymaker
    - other

Application categories may include:

- student tutoring / personalized learning
- teacher support
- tutor support
- assessment
- feedback
- prediction / early warning
- accessibility
- language / translation
- administration / system management

E. Intervention design

Extract:

- What exactly was introduced?
- Who received the intervention?
- What did users do with the AI?
- What role did teachers or other humans play?
- Did AI:
    - replace instruction
    - supplement instruction

    - support a teacher/tutor
    - provide feedback
    - make predictions
    - support administration
- Duration
- Frequency
- Session length
- Intensity
- Location of use
- Control/comparison condition, if applicable

H. Implementation

Implementation Facilitators = factors that helped the organization deliver the intervention as intended.

Extract concrete information on:

- teacher training
- staff training
- technical support
- monitoring
- implementation staff
- implementation fidelity
- incentives
- teacher involvement
- human-AI interaction
- workflow integration
- organizational requirements
- digital literacy
- technical difficulties
- resistance
- trust
- privacy concerns
- data requirements
- internet problems
- electricity problems
- device constraints
- other implementation barriers
- implementation facilitators

I. Deployment

Extract:

- school / home / community / remote
- classroom / computer lab / individual device
- smartphone
- tablet
- laptop
- desktop
- basic mobile phone
- SMS
- WhatsApp
- web platform
- mobile app
- internet requirement
- offline capability
- electricity requirement
- language
- localization
- curriculum alignment
- integration with existing school systems
- integration with government systems
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J. Scaling

Extract factual information on:

- initial pilot size
- later deployment size
- number of students
- number of teachers
- number of schools
- number of districts
- number of regions
- geographic expansion
- expansion over time
- whether government adopted the tool
- whether the tool was integrated into public schools
- institutionalization
- procurement
- modifications made when scaling
- staffing changes
- infrastructure changes
- operational requirements at scale

Do not label something "scaled" only because the authors describe it as scalable. Record actual deployment numbers where possible.

K. Cost

Extract:

- total program cost
- cost per student
- cost per teacher
- cost per school
- cost per user
- marginal cost
- fixed cost
- software cost
- hardware/device cost
- connectivity cost
- training cost
- staffing cost
- technical-support cost
- infrastructure cost
- cost-effectiveness measure
- period over which cost is calculated
- currency
- year of cost estimate

Do not combine marginal cost with total cost.

L. Financing

Extract:

- who paid for development
- who paid for the pilot
- who paid for implementation
- who paid for scale-up
- government funding
- school funding
- household/user fees
- donor funding
- philanthropy
- NGO funding
- development-bank funding
- private investment
- subscription
- procurement
- grant funding
- public-private partnership

- results-based financing
- outcome-based financing
- development impact bond
- other financing mechanism

When possible, distinguish:

- research funding
- pilot financing
- operating financing
- scale-up financing

Do not infer a financing source unless explicitly stated.

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8. Missing information

Do not guess.

If a field is unavailable, record:

Not reported

If information was searched for but could not be verified, record:

Not verified

Do not infer values from surrounding information.

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9. Output format

Produce a structured evidence table. Use one row per intervention/deployment, not necessarily one row per paper.

For each source, produce two separate sheets. Sheet 1 (General sheet) contains the following columns; Sheet 2 (Outcome sheet) follows the Education Intervention Outcome Extraction Rule. Use matching Study / Source and AI Tool labels to link the sheets, and distinguish multiple deployments in those labels where needed.

1. Tool ID
2. Intervention ID
3. Source ID(s)
4. Short Citation

5. Year
6. Country and City
7. Country Income Level
8. Publication Type
9. Study Design
10. AI Tool
11. AI Function Category
12. AI Function Description
13. Primary User
14. Educational Problem
15. Education Level
16. Targeted Population of Intervention
17. Sample Size
18. Implementing Organization
19. Treatment Condition
20. Control Condition
21. Teacher / Tutor Involvement
22. Duration
23. Frequency / Intensity
24. Implementation Requirements
25. Implementation Barriers
26. Implementation Facilitators
27. Deployment Model
28. Device / Infrastructure
29. Language / Localization
30. Deployment Stage
31. Scaling Information
32. Government Involvement
33. Cost Information
34. Financing Source
35. Financing Mechanism
36. Full citation
37. Official source URL

Keep each cell concise but information-rich.

Where one source provides several relevant quantitative results, include the main results rather than summarizing them only as "positive" or "negative."

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10. Final output requirements

The main output should be a **clean, verified evidence database** that can later be used for cross-study analysis.

The goal is:

> **To collect comparable factual information on how AI education tools are designed, adopted, implemented, deployed, used, scaled, funded, and associated with educational outcomes.**